

Critical defence technologies required by DRDO through offset partners

GUNS

- Barrel technology
- Advanced recoil system
- EM rail gun technology
- Muzzle reference system

AMMUNITION

- FSAPDS (Fin Stabilised Armour Piercing Discarding Sabot)
- HESH (High Explosive Squash Head)
- Surface-coated double-base propellant
- Composite sabot manufacturing technology

RADAR/RF

- Miniature SAR and ISAR
- Shared and conformal apertures
- PPN technology
- THZ technology

LASER

- Fibre-laser technology
- High-power lasers

MISCELLANEOUS

- Solar-cell technology
- Super-cavitation technology
- Low-observation technology
- Molecularly-imprinted polymers
- Carbon-fibre technology
- MEMS-based sensors, actuators, RF devices and focal plane arrays
- Nanotechnology-based sensors and displays

OEM to discharge offsets. Col. K.V. Kuber, Indian Army veteran advisor, NSIC (National Small Industries Corporation), explains, "Offsets are not grace, charity or a discount; they are part of an industrial participation programme, and must make business sense to the OEM and the local partner."

Robert John, CEO, Amphenol Interconnect, and past president, ELCINA, gives a deeper insight into becoming an offset partner. "I would like to tell the many companies that have not yet become an offset partner, and are trying to get a licence from the MoD, that it took us two-and-a-half years. We overcame the first challenge and got the offset partner validity, but not the licence. We could provide offset to people who wanted it in the country.

But for two years, till 2012, we got no business whatsoever, though we signed several MoUs with foreign OEMs," he says.

Opportunities in the modernisation of the Defence forces

"India is amongst the top ten countries in the world when it comes to Defence expenditure. India's current Defence budget for 2013-14 is ₹ 2036.7 billion, of which approximately 40 per cent goes into capital procurement," informs Lt Gen A.K.S. Chande, PVSM, AVSM (DG, EME ret'd). Electronics pervades every sphere of Defence equipment. The next decade is likely to see an exponential growth in combat systems as well as non-platform based Defence strategic electronic systems.

"We have seen an increase in demand for hand-held T&M devices in the Defence and aerospace industry. Nowadays, almost the same specifications of bench-top products are available in the hand-held/portable form factor. Big bulky vector network analysers or vector voltmeters are

being replaced by hand-held vector network analysers. So it is now easy to carry them to the instrument landing system (ILS) in the flight cockpit, which would allow cable parameter measurement inside a plane or ship," explains Madhukar Tripathi, regional manager, Anritsu India Pvt Ltd.

"Defence acquisition planning is a very structured process. The three services (Army, Navy and Air Force) get together to create a LTIPP (long term integrated perspective plan), currently in a 15-year cycle starting from 2012, going up to 2027," explains Chande. He adds, "From this LTIPP, five-year plans are derived that run concurrent to the national five-year Plan. There is a firm com-

mitment of funds allocated to this plan. From this five-year plan, an annual acquisition plan is drawn out. Based on the LTIPP, the government had shared with the industry a technology perspective in April this year, called the Technology Perspective Capability Road Map."

The Long Term Integrated Perspective Plan (2012-2027). The major components of the LTIPP for all the three services are shown in the box on this page. For the Army, the Tactical Communication Systems (TCS) is currently under development and has two contenders - BE (Bharat Electronics) and a consortium of L&T, TCS and HCL. Network Centric Warfare (NCW) has a number of elements that are being developed separately. The Battlefield Management System (BMS) is a pro-

Long Term Integrated Perspective Plan (2012-2027)

ARMY

- Tactical Communication Systems (TCS), Network Centric Warfare (NCW), Battlefield Management Systems (BMS), Future Infantry Soldier As A System (FINSAS), Electronic Warfare (EW) systems, air defence and tank electronics, radar, sensors and night-vision devices

NAVY

- Acquisition of a variety of surface ships, submarines and aircraft
- 46 vessels under order, including two indigenous aircraft carriers
- Apart from six Scorpene Class submarines, six more diesel and three to five nuclear submarines are to be acquired

AIR FORCE

- Procurement of 126 MMRCAs, transport aircraft (Hercules, Globemaster), AEW aircraft, trainers, additional Sukhois, and upgradation of the Mirage
- Jaguar, MIG-29

INTERNAL SECURITY

- Surveillance and night-vision devices
- Communication equipment

COMMON

- Helicopters (heavy-lift, attack, utility) and UAVs (including UCAVs) for all three Defence services and Central Armed Police Forces